MIL-C-43519A

7 December 1971
SUPERSEDING
MIL-C-43519
3 August 1967

MILITARY SPECIFICATION

CONVEYOR, STORAGE AND RETRIEVAL, DRY-CLEANED-GARMENT, MOTORIZED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers motorized storage and retrieval conveyors for handling dry-cleaned garments on coat hangers.

1.2 Classification.- The storage and retrieval conveyors shall be 50 inches wide overall including garments, and of the following overall lengths as specified (see 6.2):

Length - 14 feet
Length - 20 feet
Length - 26 feet
Length - 30 feet
Length - 36 feet
Length - As specified*

* NOTE: Users should specify standard size conveyors only unless existing operating areas necessitate a change.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-636 - Motors, Alternating-Current, (Fractional Horsepower).

FSC 3510

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MIL-L-3153 - Laundry and Dry Cleaning Machinery and Equipment (For Fixed Installation) Preparation for Delivery Of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Electrical Manufacturer's Association (NEMA) Standards Publication

No. ICS - Industrial Controls and Systems (1970)

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, N.Y. 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - National Electrical Code (1968)

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

Underwriters' Laboratories, Inc. (UL)

Electrical Construction Materials List

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Ill. 60611; 1285 Walt Whitman Road, Melville, Long Island, N.Y. 11746; or 1655 Scott Boulevard, Santa Clara, Calif. 95050.)

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3. REQUIREMENTS

* 3.1 First article.- When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.2 and 6.5).

3.2 Standard product.- The storage and retrieval garment conveyors covered by this specification shall be the regular commercial product of the manufacturer except for changes necessary to comply with the specification requirements.

* 3.3 Code and standards.-

3.3.1 NFPA.- The installation of the electrical components of the conveyor shall conform to the requirements of NFPA Standard No. 70 - National Electrical Code for general use.

3.3.2 UL.- The electrical materials and components shall comply with UL standards, as applicable.

3.3.3 NEMA.- Motor controllers shall conform to performance requirements of NEMA Standards Publication No. ICS-1970.

3.3.4 Certification.- Prior to approval of the first article or if none is submitted, prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative, that the electrical materials and components he proposes to furnish meet the requirements specified in 3.3.2 and 3.3.3, and that the components are installed as specified in 3.3.1.

3.3.4.1 NFPA.- Acceptable evidence of meeting requirements of 3.3.1 shall be the manufacturer's certified statement that the installation of the electrical components conforms to NFPA Standard No. 70.

3.3.4.2 UL.- Acceptable evidence of meeting requirements of 3.3.2 shall be the UL label or Listing Mark or certification from an independent testing laboratory acceptable to the contracting officer or his authorized representative that the electrical materials and components conform to applicable UL standards.

3.3.4.3 NEMA.- Acceptable evidence of meeting the requirements of 3.3.3 shall be the manufacturer's certified statement that motor controllers conform to NEMA Standard Publication No. ICS-1970. A tag or label attached to the controller(s) stating the component conforms to this standard is acceptable evidence.

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* 3.4 Materials.- Materials shall be as specified hereinafter. Materials not definitely specified shall be of a quality suitable for the purpose intended, provided the completed item complies with all the provisions of this specification.

3.5 Design and construction.- Each garment conveyor shall be electrically operated and equipped with rails of the nonslick type, or link and load bar type around the perimeter of conveyor, designed for storing and automatically delivering garments to a pre-selected location. The height of the conveyors shall be adjustable from 74 inches to 80 inches or shall be a fixed height as specified (see 6.2). The overall lengths of the conveyor shall be 14 feet, 20 feet, 26 feet, 30 feet, and 36 feet, plus or minus 6-inches, or other lengths as specified (see 6.2). The overall width shall be 50 inches, plus or minus 2 inches. Length and width dimensions shall include garments. When specified, the conveyor shall conform to the configuration required and in addition, when specified shall have double decked units (see 6.2). The conveyors shall be floor mounted and shall be capable of supporting the loads specified in table I. When functionally and operationally tested as specified in 4.4.1 and 4.4.2, the completely assembled conveyor shall show that the assembly of the components have:

(a) Complete and proper interrelationship.
(b) Motor bearings show no evidence of overheating.
(c) Starting and stopping devices operating smoothly and positively.
(d) No structural failures.

TABLE I.- Capacity

Length	Capacity (minimum)
14 Feet	1800 Pounds
20 Feet	2600 Pounds
26 Feet	3400 Pounds
30 Feet	4000 Pounds
36 Feet	4800 Pounds
* For longer conveyor lengths as specified	125 pounds per foot in length minimum

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3.5.1 Frame.- The frame supporting the drive and rail shall have sufficient adjustable supporting columns to meet the requirements specified in 3.5. The rails shall form a part of the frame and shall be of steel. The load carrier shall be suspended from the rail by nylon tired, ball bearing rollers, and shall contain notches or slots to hold the garment hangers to prevent the hangers from sliding when in operation. The traveling load carriers shall contain an identification number or letter (see 3.5.4.1) for each notch or slot. The frame base shall be provided with sufficient anchor bolt holes for anchoring the assembled conveyor to the floor. When tested as specified in 4.4.3, the anchored frame base shall not tip when supporting an unbalanced load. Means shall be provided to allow adjustment for leveling the frame.

3.5.2 Drive.- The conveyor shall be electric motor driven through a V-belt and pulleys, gears, or roller chain and sprockets. Motor horsepower shall be capable of driving the conveyor loaded to maximum capacity, at a minimum rated speed of 72 feet per minute (FPM), when tested as specified in 4.4.1 and 4.4.2.

3.5.3 Electrical characteristics.- Unless otherwise specified (see 6.2), the conveyor shall operate from a 120 volt, 60 hertz, 1-phase power source. Motors shall be in accordance with CC-M-636.

3.5.4 Controls.- Unless otherwise specified (see 6.2), each conveyor shall be equipped with both manual and automatic electric controls.

* 3.5.4.1 Automatic control.- The automatic control shall be of the remote type. The control shall have a dial or push-button system with numerals or letters, corresponding with the numerals or letters on the load carrier. When the control is dialed or a push-button pressed for any certain numeral or letter, the control shall activate a switch and the drive motor shall cause the conveyor to rotate in a direction of shortest travel and accurately stop the load carrier with the corresponding number or letter at the established delivery station. The control shall also permit the conveyor to rotate and operate continually without a stop at delivery station.

3.5.4.2 Manual control.- Each conveyor shall be equipped with a pair of foot switches (for right and left operation). When stepped on, the foot switches shall cause the conveyor to revolve in either direction, depending upon the switch pressed, and cause the conveyor to stop immediately when the switch is released.

3.6 Finish.- Unless otherwise specified (see 6.2), the conveyor shall be finished in accordance with the manufacturer's standard practice.

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3.6.1 Application of finish.- The finish applied to the end item shall be continuous, smooth, adherent without discoloration or foreign material imbedded and contain no sags, runs, drips, creeps, laps, bubbles, cracks, pours, pits, lumps, flux, or orange peel. No rust, rough grinds, or toolmarks shall show through the coating.

3.7 Marking for identification.- Each conveyor shall be marked for identification in accordance with MIL-STD-130.

3.8 Electromagnetic compatibility.- When specified (see 6.2), equipment shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.4).

3.9 Workmanship.- The conveyor shall be free from defects such as fractures, splits, or dents. There shall be no sharp edges, burrs, or slivers.

3.9.1 Welding.- Welds shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits shall be removed from finished welds.

3.9.2 Bolts, nuts, screws, studs and other threaded fasteners.- Threaded fasteners shall not be broken, stripped, fractured, or loose.

3.9.3 Driving of rivets.- Rivets shall be driven to completely fill the holes, with full size heads neatly finished and in full contact with the surface of the members.

3.9.4 Electrical wiring.- Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals. Wiring shall have adequate slack to provide strain relief.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.- When a first article is required, (see 6.5) it shall be examined for defects in table II, specified dimensions, and tested as specified in 4.4.1, 4.4.2, 4.4.3 and when required 4.4.4. The presence of any visual defect, any dimension not within the specified requirements, or failure to pass the tests shall be cause for rejection of the first article.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 End item inspection.- An inspection lot shall consist of all conveyors of one length offered for inspection at one time. The sample unit shall be one completely fabricated conveyor.

4.3.2.1 Visual examination.- Examination shall be made of the conveyor for defects in table II. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Finish not as specified	X	
	Finish not continuous, smooth, adherent, discolored, contains sags, drips, creeps, laps, bubbles, imbedded or foreign material, cracks, pours, pits, flux or orange peels, rust, rough grinds or toolmarks showing through finish		X
Construction	Any part missing or not specified type or size	X	
	Any component split, fractured, dented, or malformed	X	
	Component misplaced or not in proper alignment	X	

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TABLE II.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction (cont'd)	Component loose		X
	Sharp burr, sliver, or sharp edge	X	
	Any component that is inoperative or will not function as intended	X	
	Any component that requires abnormal force to operate	X	
	Component not connected or joined properly	X	
Welding and brazing	Missing, incomplete, burn holes, fractured, or not fused	X	
	Slag inclusion, undercut, not smooth and uniform, or deposits of scale and flux removed		X
Electrical assembly	Wiring cut, abraded	X	
	Adequate slack not provided for wiring or excessive insulation stripped from wiring		X
Bolts, nuts screws and other threaded fasteners	Broken, stripped or fractured	X	
	Threaded fastener loose		X
Rivets	Not tightly set, not in full contact with surface of connecting member	X	
Identification marking	Missing, incomplete, not legible, incorrect		X

4.3.2.2 Dimensional examination.- Examination shall be made of the conveyor for compliance with dimensional requirements. Any deviation from the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

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* 4.3.2.3 Testing of the end item.- When a first article is not required, the initial production unit shall be tested as specified in 4.4.2, 4.4.3 and when required 4.4.4. Any nonconformance shall be cause for rejection. Each unit of production shall be tested as specified in 4.4.1. Any nonconformance shall be cause for rejection of the unit.

4.3.3 Code and standards compliance.- Proof of compliance with the requirements of 3.3 shall be submitted to the Government representative.

4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that packaging and packing, preservation and marking conform to section 5 of this specification.

4.4 Tests.-

4.4.1 Operational testing.- The conveyor shall be connected to power and operationally tested, without any garments on hooks. In the course of testing for operational requirements specified in 3.5, determination shall be made to establish compliance with all performance requirements applicable to the fully assembled item except those that require functional testing such as operating the conveyor for load test. The conveyor shall be operated for not less than 1/2 hour to determine compliance with 3.5 and 3.5.2. Any nonconformance with operational requirements of 3.5 and 3.5.2 shall constitute failure of this test.

* 4.4.2 Functional test.- The conveyor shall be connected to power with all notches or slots loaded. The total load on the conveyor shall be as specified in table I. The loaded conveyor shall be run for 3 minutes and stopped. This test shall be repeated 6 times and the conveyor examined for conformance with the requirements of 3.5 and 3.5.2. Any nonconformance with functional requirements of 3.5, 3.5.2, 3.5.4.1 and 3.5.4.2 shall constitute failure of this test. A load equal to one-half the maximum capacity specified in table I shall be loaded to one side only of the conveyor.

4.4.3 Unbalanced load test.- The conveyor shall be started and run for 3 minutes and stopped. This test shall be repeated 6 times and the conveyor shall be examined for conformance with 3.5.1. Any tilting or tipping shall constitute failure of this test.

* 4.4.4 Electromagnetic compatibility.- When electromagnetic compatibility is required, the first article or initial unit of production, as applicable, shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the Contracting Office written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromganetic test report and the requirements meet MIL-STD-461.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging, packing, and marking.- Preservation and packaging, packing, and marking for level A, B, or C, shall be in accordance with MIL-L-3153 (see 6.2).

6. NOTES

6.1 Intended use.- The garment conveyors covered by this specification are intended for use in dry cleaning call offices for storing and delivery of garments on hangers to an established station for ready disbursement by the operator.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Length of conveyor required (see 1.2 and 3.5).
(c) When a first article is required (see 3.1, 4.2 and 6.5).
(d) When fixed height is required (see 3.5).
(e) When a specified configuration of conveyor and when double decked units are required (see 3.5).
(f) Electrical characteristics, if other than specified (see 3.5.3).
(g) When controls are to be other than specified (see 3.5.4).
(h) Finish, when other than specified (see 3.6).
(i) When electromagnetic compatibility is required (see 3.8).
(j) Selection of applicable levels of preservation and packaging and packing (see 5.1).

* 6.2.1 Contract data requirements.- Any requirements for equipment manuals for the items covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

6.3 Functional testing.- The functional tests (4.4.2 and 4.4.3) required in 4.3.2.3 for the first unit of production may be waived by the contracting officer, if this test has been performed previously on an identical conveyor and the supplier provided certified test data indicating compliance with the requirements.

6.4 Successful commercial operation.- Invitation for bids should specify that no item of equipment will be acceptable unless the manufacturer has had equipment of approximately the same design as that specified, operating successfully in a commercial or institutional laundry or laundry call office for at least 1 year. Equipment installed for test purposes in a manufacturer's plant or laboratory shall not come within the category of successful commercial operation.

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* 6.5 First article.- When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should consist of one preproduction sample of the size conveyor specified. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

* 6.6 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 84

Preparing activity:

Army - GL

Project No. 3510-0193

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
CONVEYOR, STORAGE AND RETRIEVAL, DRY-CLEANED-GARMENT, MOTORIZED MIL-C-43519A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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